

Additional text 7: does complex membership explain the heat response?

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Question

Measured thermal stability explains only about 1 % of the 43 vs 35 °C response (additional text 1). One hypothesis for why: heat disassembles protein complexes, and the released subunits become DNAJ clients whatever their own stability. It predicts that (1) at matched stability, complex subunits gain more co-chaperone association than other proteins, (2) subunits of the same complex move together, and (3) both effects are weaker at 37 °C, where little is perturbed.

Data

Human complexes from the EBI Complex Portal (curated; 2,167 complexes with at least two protein subunits after resolving sub-complexes; 3,478 proteins). CORUM, a larger catalogue, could not be downloaded. 2,471 of the 8,217 GA_33 proteins with a heat response are subunits of at least one complex. Meltome melting temperatures as in additional text 1.

Results

1. Complex subunits gain more at 43 °C, at matched stability. Difference in the 43 vs 35 °C response, subunits minus other proteins, adjusted for melting temperature (quadratic), level in the 35 °C pull-down (quadratic) and nuclear, cytoplasmic and membrane location; 95 % interval over sequence clusters:

	Raw	Adjusted
Heat 43 °C	+0.11 (p 7e-11)	+0.17 [+0.12 , +0.22]
Heat 37 °C	+0.03 (p 5e-8)	+0.04 [+0.02, +0.05]

The effect is heat-dependent, as predicted. It is concentrated at intermediate stability:

Melting temperature third	Subunits	Other proteins
Least stable	+0.21 (812)	+0.30 (1,563)
Middle	+0.45 (796)	+0.28 (1,585)
Most stable	+0.15 (671)	+0.12 (1,704)

Subunits show the stability window of additional text 1 more sharply than other proteins: they gain most when moderately stable and least when least stable. But membership does not account for the window itself; the curvature in melting temperature is unchanged when membership is added to the model (-0.063 against -0.067).

2. Subunits move together, but at both temperatures. The responses of a complex’s detected subunits are more alike than those of random sets of the same size (804 complexes with at least three detected subunits):

Share of the response variance shared within a complex	43 °C	37 °C
Against random proteins	37 %	36 %
Against random subunits of other complexes	27 %	29 %
Same, after removing level, Tm and location	24 % / 14 %	36 % / 31 %

$p \leq 0.002$ throughout. Coherence is as strong at 37 °C as at 43 °C, so it is not a heat effect. Its most likely source is that pull-downs capture stable complexes as units: when the co-chaperone holds one subunit, its partners come along. Once level and location are removed, coherence relative to other complexes’ subunits is lower at 43 °C (14 %) than at 37 °C (31 %), which is what partial dissociation of complexes by heat would produce, but this comparison alone does not establish it.

3. Which complexes move. Mean 43 vs 35 °C response of the detected subunits:

Gain most		Lose most	
eIF3 translation initiation factor (13)	+2.71	Retention and splicing complex (3)	-1.73
RNase H2 (3)	+2.68	CDK-activating kinase (3)	-1.57
COPII vesicle coat (8)	+2.53	Chromosomal passenger complex (4)	-1.13
COP9 signalosome (8)	+2.49	Nucleosome (3)	-0.97
GIN5 (3)	+2.45		

Cytoplasmic machinery for translation initiation, vesicle traffic and protein degradation gains most; chromatin-bound nuclear complexes lose most, in line with the class effects found earlier.

Interpretation

Complex membership matters in the direction the hypothesis predicts. Subunits gain more co-chaperone association with heat at matched stability, abundance and location, and do so most when moderately stable. But the effect is small. A +0.17 shift carried by 30 % of proteins accounts for well under 1 % of the variance of a response whose spread is 1.1 log2 units, comparable to melting temperature. Neither stability nor complex membership explains much of the heat response; protein class and the sequence model explain far more.

Critical evaluation

- **Catalogue coverage.** Complex Portal is curated but incomplete; many real complexes are missing, so some “non-members” are subunits. That dilutes the membership effect rather than creating it.
- **Disassembly is inferred, not observed.** Pull-downs do not show whether a complex is intact. The lower coherence at 43 °C is suggestive only.
- **Coherence has a technical explanation.** Co-purification of intact complexes produces coherence at any temperature, which fits the 37 °C result.
- **The input lysate** would show whether gaining subunits are released from complexes (constant total level) or newly abundant.

Methods files

code/ad07_complexes.py; results in data/ga_data/ad07_results.json.